One differential equation, two transforms: a proof of Layman's conjecture on OEIS A005329

Adrian Perez Fontelles

Independent researcher

24 August 2026

Abstract

Let $a(n) = \prod_{i=1}^{n}(2^i - 1)$ be the 2-factorial numbers, OEIS `A005329`. In September 2002 J. W. Layman conjectured that this sequence is the inverse binomial transform of `A075272`, the BinomialMean transform of `A075271`. We prove it. Both sides are governed by a functional equation for an exponential generating function, and the two equations are carried into one another by the substitution $g = e^x f$ — which is precisely the operation "take the binomial transform". On the `A005329` side the equation is nothing but the defining recurrence $a(n+1) = (2^{n+1} - 1)a(n)$ rewritten; on the `A075272` side it is the definition of the BinomialMean transform rewritten. The conjecture is the observation that these two rewritings differ by exactly one factor of e^x.

1 The sequences and the conjecture

OEIS `A005329` has offset 0 and is

$$a(n) = \prod_{i=1}^{n}\bigl(2^i - 1\bigr) = 1,\ 1,\ 3,\ 21,\ 315,\ 9765,\ 615195,\ 78129765, \ldots$$

equivalently $a(0) = 1$ and

$$a(n+1) = \bigl(2^{n+1} - 1\bigr)a(n) \qquad (n \geq 0). \tag{1}$$

The *BinomialMean transform* BM of a sequence $(u(n))$ is

$$(\mathrm{BM}u)(n) = \frac{1}{2^n}\sum_{k=0}^{n}\binom{n}{k}u(k),$$

and `A075271` is the sequence b defined by $b(0) = 1$ together with

$$(\mathrm{BM}b)(n) = 2\,b(n-1) \qquad (n \geq 1), \tag{2}$$

which determines b uniquely; `A075272` is $c = \mathrm{BM}b$. Thus $b = 1, 3, 17, 211, 5793, \ldots$ and $c = 1, 2, 6, 34, 422, 11586, \ldots$, and (2) says

$$c(n) = 2\,b(n-1) \qquad (n \geq 1), \tag{3}$$

a relation also recorded as a comment on `A075272`.

The entry `A005329` carries

Conjecture: this sequence is the inverse binomial transform of `A075272` *or, equivalently, the inverse binomial transform of the BinomialMean transform of* `A075271`*. — John W. Layman, Sep 12 2002*

(The two readings coincide, since `A075272` $=$ BM(`A075271`) by definition.) As of the "Last modified" line on the live entry (24 August 2026) the statement is still recorded as an unproven conjecture, with no proof in the entry's links or programs. Writing $\mathcal{B}$ for the binomial transform, $(\mathcal{B}u)(n) = \sum_k \binom{n}{k} u(k)$, whose inverse is $(\mathcal{B}^{-1}u)(n) = \sum_k (-1)^{n-k}\binom{n}{k}u(k)$, the conjecture says

$$\mathcal{B}a = c. \tag{4}$$

Remark (status, stated plainly). A comment added to `A075272` by V. Reshetnikov on 20 November 2015 asserts "Binomial transform of `A005329`" flatly, without label or proof, and the Mathematica program on that entry generates the terms from that identity. So the same statement sits on one entry as an unproven conjecture and on the other as an unsourced assertion, and the two have never been reconciled. I have found no proof on either entry or in the literature. What follows supplies one.

2 Two functional equations

All generating functions are formal exponential generating functions over $\mathbb{Q}$; all manipulations are formal.

Lemma 1. *Let* $f(x) = \sum_{n\geq 0} a(n)x^n/n!$*. Then* (1) *holds if and only if*

$$f(x) + f'(x) = 2f(2x). \tag{5}$$

Proof. The coefficient of $x^n/n!$ in f is $a(n)$, in f' is $a(n+1)$, and in $2f(2x)$ is $2\cdot 2^n a(n) = 2^{n+1}a(n)$. So (5) is equivalent to $a(n) + a(n+1) = 2^{n+1}a(n)$ for all $n \geq 0$, which is (1). $\square$

Lemma 2. *The sequence* $c =$ `A075272` *is the unique sequence with* $c(0) = 1$ *satisfying*

$$\sum_{k=0}^{n}\binom{n}{k}c(k+1) = 2^{n+1}c(n) \qquad (n \geq 0). \tag{6}$$

Moreover, if $g(x) = \sum_{n\geq 0} c(n)x^n/n!$*, then* (6) *holds if and only if*

$$e^x g'(x) = 2g(2x). \tag{7}$$

Proof. By (3), $b(k) = \frac{1}{2}c(k+1)$ for $k \geq 0$. Substituting this into $c = \mathrm{BM}b$, that is into $2^n c(n) = \sum_{k=0}^n \binom{n}{k}b(k)$, gives $2^n c(n) = \frac{1}{2}\sum_{k=0}^n \binom{n}{k}c(k+1)$, which is (6). Also $c(0) = (\mathrm{BM}b)(0) = b(0) = 1$. Conversely (6) determines $c(n+1)$ from $c(0), \ldots, c(n)$, since the $k = n$ term is $c(n+1)$ and all other terms involve smaller indices; with $c(0) = 1$ this fixes the whole sequence, giving uniqueness.

For the second claim: the binomial transform corresponds to multiplication of exponential generating functions by e^x, so $\sum_n \big(\sum_k \binom{n}{k}c(k+1)\big)x^n/n! = e^x g'(x)$, while $\sum_n 2^{n+1}c(n)x^n/n! = 2g(2x)$. $\square$

3 The theorem

Theorem 3. *$\mathcal{B}a = c$; that is, for every $n \geq 0$,*

$$\sum_{k=0}^{n} \binom{n}{k} \prod_{i=1}^{k} (2^i - 1) = \texttt{A075272}(n), \qquad \textit{equivalently} \qquad a(n) = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} \texttt{A075272}(k).$$

Proof. Let f be the exponential generating function of a and put $g := e^x f$, which is the exponential generating function of $\mathcal{B}a$. Then

$$e^x g' = e^x \big(e^x f + e^x f'\big) = e^{2x}\big(f + f'\big), \qquad 2g(2x) = 2e^{2x} f(2x).$$

By Lemma 1, $f + f' = 2f(2x)$, so the two right-hand sides agree and g satisfies (7). Also $(\mathcal{B}a)(0) = a(0) = 1$. By Lemma 2, g is the exponential generating function of A075272, whence $\mathcal{B}a = c$. Applying $\mathcal{B}^{-1}$ gives the second form, which is Layman's conjecture. □

Remark (what the proof is really saying). Both sequences are pinned down by a single functional equation, and the two equations differ by exactly one factor of e^x — which is what the binomial transform does to an exponential generating function. So the conjecture is not an accident of the numbers: it says that the recurrence defining the 2-factorials and the recurrence defining A075271 through its BinomialMean transform are the same statement, one binomial transform apart. The factor 2 in (2) is what matches the base 2 in $\prod(2^i - 1)$; the same argument with q in place of 2 relates $\prod_{i=1}^{n}(q^i - 1)$ to the analogous q-mean transform.

Remark (honest assessment). The proof is half a page and uses nothing beyond formal exponential generating functions. It is an OEIS comment, not a paper. It is worth recording because the statement has carried the label "Conjecture" since 2002, because the entry A075272 has meanwhile asserted the same identity as a fact since 2015 without proof, and because the two entries have never been connected.

4 Verification

A verification script, written separately from this note, checks every claim and prints every number quoted. It (i) reproduces the first 13 published terms of A005329 from the product definition; (ii) generates A075271 from its definition (2), obtaining $1, 3, 17, 211, 5793, 339491, 41326513, \ldots$ forms $c = \mathrm{BM}b$, obtaining $1, 2, 6, 34, 422, 11586, 678982, \ldots$ in agreement with the published DATA of A075272, and confirms (3) for $n \leq 24$; (iii) confirms Layman's conjecture directly, checking $\mathcal{B}^{-1}(\texttt{A075272})(n) = \texttt{A005329}(n)$ and equivalently $\mathcal{B}(\texttt{A005329})(n) = \texttt{A075272}(n)$ for $n = 0, \ldots, 24$, with no discrepancy; (iv) confirms the coefficient forms of Lemmas 1 and 2, namely $a(n) + a(n+1) = 2^{n+1} a(n)$ and $\sum_k \binom{n}{k} c(k+1) = 2^{n+1} c(n)$, for $n \leq 23$; (v) confirms that $\mathcal{B}a$ itself satisfies (6), which is the content of Theorem 3 at the level of coefficients; and (vi) confirms the uniqueness claim of Lemma 2 by regenerating A075272 from (6) and $c(0) = 1$ alone.

References

[1] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org/A005329, consulted 24 August 2026; also A075271 and A075272.

[2] M. Z. Spivey and L. L. Steil, *The k-binomial transforms and the Hankel transform*, Journal of Integer Sequences **9** (2006), Article 06.1.1.

A Suggested OEIS comment (plain text, house style)

```
Layman's conjecture is true. Write B for the binomial transform, so that the
claim is B(A005329) = A075272, and let f and g be the e.g.f.s of A005329 and
A075272. Two rewritings:

(a) The defining recurrence a(n+1) = (2^(n+1) - 1) a(n) of A005329 says
    a(n) + a(n+1) = 2^(n+1) a(n), i.e. f + f' = 2 f(2x).

(b) A075272 = BM(A075271) and A075272(n) = 2*A075271(n-1) together give
    A075271(k) = A075272(k+1)/2; substituting into 2^n*A075272(n) =
    Sum_k C(n,k) A075271(k) yields Sum_k C(n,k) c(k+1) = 2^(n+1) c(n),
    i.e. e^x g' = 2 g(2x). With c(0) = 1 this determines A075272 uniquely.

Now the binomial transform multiplies an e.g.f. by e^x, so put g = e^x f.
Then e^x g' = e^x (e^x f + e^x f') = e^(2x)(f + f') and 2 g(2x) =
2 e^(2x) f(2x), so (a) gives exactly e^x g' = 2 g(2x), which is (b). Since
also (B a)(0) = a(0) = 1, uniqueness in (b) gives B(A005329) = A075272, i.e.
A005329 is the inverse binomial transform of A075272.

The two equations differ by a single factor of e^x, which is precisely what
the binomial transform does; that is the whole content of the conjecture.
Note that A075272 has carried the unlabelled comment "Binomial transform of
A005329" (V. Reshetnikov, Nov 20 2015) while this entry has carried the same
statement as a conjecture since 2002; the above proves both.
```